DSIP

* Assignment - 0

Q-1] (a) $\{1, 2, 1, 2, 3\}$

z-transform : $x(z) = 1 + 2z^{-1} + 1z^{-2} + 2z^{-3} + 3z^{4}$

ROC : Entire z-plane
causal : yes (strat from $n = 0$)
Stable : yes

b] $x(n) = (n+1)^2 u(n)$

z-transform $x(z) = \sum_{n=0}^{\infty} (n+1)^2 z^{-n}$

ROC : $|z| > 1$
causal : yes
Stable : No

c] $x(n) = \log(n)$

z-transform : Not defined at $n = 0$
causal : No
Stable : No

d] $x(n) = 2^n u(n) - 0.5^n u(n-1)$

z-trans $x(z) = \frac{1}{1-2z^{-1}}$; $|z| > 2$ (Right)

$+ \frac{z}{z-0.5}$, $|z| < 0.5$ (left)

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ROC : No overlapping ROC -> z - transform : undefined.

Causal : No
Stable : No

e) $x(n) = u(n-6)$
This is 1 for $n \leq -6$
$$x(z) = \sum_{n=-\infty}^{-6} z^{-n} = \frac{z^{6}}{1-z} \; ; |z| < 1$$

f) $x(n) = \delta(n+5)$
z - transform $x(z) = z^{5}$
ROC = Entire z-plane
causal : No
Stable : Yes

Q-2] Inverse z - transform
(a) $H(z) = 1 + z + z^{-1}$
$h(n) = \delta(n) + \delta(n-1)$

b] $H(z) = \log(1+z^{-1})$
$$h(n) = \frac{(-1)^{n+1}}{n} \quad n \geq 1$$

c] $H(z) = z \cdot e^{\wedge}(2z^{-3})$
$$e^{2z^{-3}} = \sum_{k=0}^{\infty} \frac{(2z^{-3})^{k}}{k!}$$ multiply by z -

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d) $H(z) = 1/((1-4z^{-1})(1-2z^{-1}))$

$H(z) = \frac{A}{1-4z^{-1}} + \frac{B}{1-2z^{-1}} \Rightarrow h(n) = A4^n + B\cdot 2^n$; $n \geq 0$

e) $H(z) = 1/((1-3)^n\, n(-1))$

This is undefined

Q-3) $y(n) = y(n-1) + 5x(n) + 2x(n-1)$

Impulse Response $h(0) = 5$, $h(1) = h(0) + 2 = 7$...

Frequency Response (z-domain)

$H(z) = \frac{5 + 2z^{-1}}{1 - z^{-1}} \Rightarrow H(e^{j\omega}) = \frac{5 + 2e^{-j\omega}}{1 - e^{-j\omega}}$

magnitude and phase : Use magnitude & angle formulas

output for $x(n) = 2^n u(n)$: $y(z) = H(z)\cdot x(z)$, then inverse z-transform

Q-4) $H(z) = 4/(z^2 - 3)$

$H(z) = \frac{4}{(z-\sqrt{3})(z+\sqrt{3})}$

Partial fraction $\Rightarrow$ inverse z-transform gives $h(n)$

Unit step response : Convolve $h(n)$ with $u(n)$

Q-5] Convolution via Z. Transform

$x(n) = 2^n u(n) \Rightarrow x(z) = \frac{1}{1-2z^{-1}}, |z| > 2$

$h(n) = .5^n u(n-1) \Rightarrow H(z) = \frac{z}{z-5}; |z| < 5$

· No overlapping ROC → Use time domain convolution

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)$$

Q-6] DTFT's

(a) $(n+1)a^n u(n)$

$$x(e^{j\omega}) = \frac{1 - ae^{-j\omega}}{(1-ae^{-j\omega})^2}, \quad |a| < 1$$

b) $\sin(\omega n)/\pi n$

Sinc function + Rectangular Frequency Resp.

c) $\cos(\omega n + B)$

DTFT contains two impulses at $\pm\omega$

d) $e^{(j\omega n)} \cdot u(n)$

$$x(e^{\Omega}) = \frac{1}{1 - e^{j(\Omega - \omega)}}$$

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Q-7] IDTFT

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

Q-8) Linear & circular convolution
compute :
linear : by time domain simmulation
circular : by wrapping modulo N.

Q-9] $(H(e^{j\omega}))\cdot 1 = 1$

Use phase to find
$h(n) = \delta(n-k)$ shifted impulse

output for $x(n) = \cos(\pi n) u(n)$
$y(n) = (\cos(\pi \times k)) u(n-k)$

Q-10] Properties of z-transform

1. Differentiation $nx(n) \leftrightarrow -z \frac{dx(z)}{dz}$
2. Time shifting $x(n-k) \leftrightarrow z^{-k} X(z)$
3. Convolution $x(n) - h(n) \leftrightarrow X(z) H(z)$
4. Time Reversal $x(-n) \leftrightarrow X(z^{-1})$

Q-11) if $x(n)$ is real then:

$$X(e^{jw}) = X^*(e^{-jw})$$

even magnitude odd phase

Q-12) $x(n) = (1,1,2,2)$ $h(n) = (1,2,3,4)$

(a) Linear from (zero pad to $N=7$ use circular conv

b) linear from circular : Wrap around sum modulo.

c) linear from DFT | IDFT : Compute DFT → Multiply → IDFT

Q-13) 8 - point DFT of $x(n) = \cos(k\pi/4)$

$x(n) = \cos(\pi/4)$ → periodic with $N = 8$

DFT = Peaks at bins 1 and 7

Amplitude spectrum = 1 at $k = 1$ & 7

Phase : 0 at $k = 1$; π at $k = 7$

Q-15) Plot $x(n) = u(n) + 3\delta(n-1) + 4u(-n-3)$

$x(n)$ = step + impulse + flipped step

$x(n-1)$ = shift right

$x(2n)$ = Compression

$x(n/2)$ = Expansion

$x^2(n)$ = Square each value

$x(-n)$ = time reversal

16) Periodicity

Signal	Periodic	Period
$e^{\pi n/6}$	✓	12
$\sin(3n)$	✓	Approx
$e^{\sin 4n}$	×	Aperiodic
$\sin(3\pi n) + \cos(6\sin)$	✓	2
$\frac{\sin(\pi n)}{\pi n}$	×	Aperiodic